When Should a Rule Learn? Predicting the Rule / Neural-Net / ML Composition for Classification Streams

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Abstract

A production classification site can be served by a hand-written rule, a (neural-net) LLM call, or a trained classical model. Common practice treats the rule→LLM→ML progression as a fixed ladder whose destination is always the trained model. We ask a data-centric question instead: **given a stream's measurable characteristics, which of these tiers should serve it, and when?** On 19 public text datasets (plus preliminary image and audio), scored apples-to-apples on a common test set across four tiers (keyword rule, zero-shot LLM, few-shot LLM, classical-ML training curve), we find a clear, interpretable gradient: **label cardinality and rule-accuracy-over-chance predict which tier wins** — high-cardinality streams favor the trained classical model, low-cardinality/binary streams favor the LLM. We operationalize this as **Adaptive Graduated Autonomy (AGA)**, which selects the *terminal* tier per stream (often the LLM, not the trained model) under a *multi-objective* policy — modality feasibility and a latency SLO as hard constraints, accuracy within a significance band, operating cost minimized — and graduates under a statistical non-inferiority gate. This paper measures the accuracy, cost, AND latency axes (modality feasibility is first-class but a hard filter, not a measured scalar). AGA matches the fixed-ladder baseline's accuracy while consuming ~**44% fewer labeled outcomes**, and keeps the LLM as terminal on roughly half of streams. We report two honest negatives that strengthen rather than weaken the picture: a *learned cross-site predictor* of the best tier does **not** generalize at this fleet size (leave-one-dataset-out below the majority baseline) — per-site measurement is required — and a vision-LLM-on-spectrogram proxy fails on environmental audio. The contribution is the characterization and the cost-reducing adaptive algorithm, not the cascade or the gate (both prior art).

1 Introduction

Most production software is full of small classification decisions: route this ticket, tag this message, flag this content, pick this branch. Each is a function from an input to one of a fixed set of labels. Teams build these three ways — a hand-written rule, a call to a large language model, or a trained classical model — and the choice is usually made once, per site, by intuition, and rarely revisited; there is no shared, evidence-based basis for it. Two progressions are well-documented in practice: replacing brittle rules with learned models (the rule-to-ML migration long noted as a source of technical debt; Sculley et al. 2015, Breck et al. 2017), and, more recently, prototyping with an LLM and distilling to a cheaper model to cut inference cost (FrugalGPT; LLM-as-teacher distillation). Both *implicitly* treat a trained model as the eventual destination — an assumption also baked into autonomy frameworks whose ceiling is a trained model. This paper tests that assumption directly rather than asserting a ladder.

This paper questions that ladder with data. For a given classification stream, which tier *should* serve it, and when? We treat it as a data-centric measurement problem: score all three tiers (rule, LLM, classical ML) on the same test set, on a shared axis of labeled-outcomes consumed, across 19 public text datasets spanning sentiment, intent, moderation, emotion, topic, and spam, plus preliminary image and audio. The answer is not "always the trained model." It depends on measurable, early-observable properties of the stream —

chiefly label cardinality and how far the day-zero rule sits above chance — and for a large fraction of streams the LLM is the correct *terminal* tier, not a way-station.

We turn this into an algorithm, Adaptive Graduated Autonomy (AGA), that chooses the terminal tier per stream under a multi-objective policy (modality feasibility and a latency SLO as hard constraints, accuracy within a statistical significance band, operating cost minimized) and graduates between tiers under a non-inferiority gate — graduating to the cheaper tier as soon as it *ties*, not only when it wins. Against the fixed "always graduate to ML" baseline, AGA matches accuracy while consuming far fewer labeled outcomes.

What is *not* claimed as novel: LLM cost-cascades (FrugalGPT) and learning-to-defer route between models already; paired-McNemar classifier comparison is standard. What *is* contributed: (1) a characterization of which dataset characteristics predict the cost-optimal tier, (2) an adaptive algorithm that exploits it to cut labeled-data cost at parity accuracy, and (3) honest evidence on where a learned cross-site policy does and does not work — including two negatives we report plainly rather than hide.

1.1 Contributions

1. **Aligned multi-tier benchmark** — rule / zero-shot LLM / few-shot LLM / classical-ML-curve, all on one common test set per dataset, on a shared labeled-outcomes-consumed axis, across 19 text datasets (sentiment, intent, moderation, emotion, topic, spam) + preliminary image/audio.
2. **Predictive characterization** — label cardinality (Spearman $\rho \approx -0.7$ vs outcomes-to-graduate) and rule-over-chance predict which tier wins and how deep ML must train.
3. **AGA** — chooses the terminal tier and graduates under a non-inferiority gate; matches fixed-ladder accuracy at ~44% fewer labeled outcomes.
4. **Two honest negatives** — a learned cross-site tier-predictor does not generalize at 21 datasets; spectrogram-vision fails on audio. Plus a tunable non-inferiority ("close-enough") gate for cost-aware graduation.

2 Setup

- **Tiers.** RULE → MODEL (LLM, zero/few-shot) → ML (classical sklearn head). The LLM is the neural-net tier; ML is the trained classical tier.
- **Datasets.** 19 text + CIFAR-10 (image) + ESC-50 (audio).
- **Aligned protocol.** Common test subset per dataset (text n=400); rule from a 100-example seed; ML as a budget curve; LLM (Claude Haiku) zero- and few-shot ($\leq$40 exemplars). Each tier placed at its true labeled-outcome cost (rule$\approx$100, zero-shot=0, few-shot=k, ML=budget).
- **Two costs.** We separate **one-time labeled-outcome cost** (to train ML / seed the rule) from **perpetual per-call inference cost** (the LLM bills every call forever; rule/trained-ML are ~free per call). AGA's savings are reported on the labeled-outcome axis; the steady-state inference-cost argument is what makes graduating *off* the LLM valuable.

3 The composition varies by stream

Sorted by label cardinality, the gradient is visible: high-cardinality → trained ML; binary/ low-cardinality → LLM. (esc50 ML "wins" only because the spectrogram-vision proxy fails; cifar10's LLM "win" reflects a weak pixel-ML baseline — see ğ6.)

dataset	labels	rule	ML	MODEL-zs	MODEL-fs	best
clinc150	151	0.172	0.838	0.748	0.725	ML
banking77	77	0.225	0.870	0.777	0.807	ML
hwu64	64	0.247	0.805	0.850	0.865	MODEL-fs
esc50	50	0.058	0.283	0.017	0.050	ML
atis	26	0.733	0.848	0.598	0.815	ML
twenty_newsgroups	20	0.133	0.660	0.685	0.680	MODEL-zs

dataset	labels	rule	ML	MODEL-zs	MODEL-fs	best
dbpedia14	14	0.632	0.968	0.968	0.978	MODEL-fs
codelangs	12	0.892	0.986	1.000	1.000	MODEL-zs
cifar10	10	0.173	0.327	0.780	0.340	MODEL-zs
yahoo_answers	10	0.240	0.670	0.740	0.720	MODEL-zs
snips	7	0.755	0.983	0.970	0.983	ML
emotion	6	0.323	0.818	0.632	0.608	ML
trec6	6	0.420	0.855	0.795	0.887	MODEL-fs
ag_news	4	0.375	0.895	0.853	0.892	ML
tweet_emotion	4	0.403	0.613	0.828	0.810	MODEL-zs
imdb	2	0.532	0.853	0.938	0.958	MODEL-fs
rotten_tomatoes	2	0.487	0.723	0.930	0.940	MODEL-fs
sms_spam	2	0.920	0.945	0.887	0.988	MODEL-fs
sst2	2	0.512	0.770	0.968	0.978	MODEL-fs
tweet_hate	2	0.497	0.560	0.770	0.640	MODEL-zs
tweet_offensive	2	0.660	0.775	0.752	0.730	ML

4 Adaptive Graduated Autonomy

- **oracle_terminal** is *multi-objective*, applied in priority order: (i) **modality feasibility** (hard filter — e.g. spectrogram-vision can't read words), (ii) **latency SLO** (hard — a tier breaching the real-time budget is out), (iii) **accuracy** within a Wilson-CI significance band, (iv) **operating cost** minimized within the band (perpetual inference \$ + labeled cost). The LLM wins only when *significantly* better on accuracy *and* feasible within the latency/modality constraints; otherwise a rule/trained-ML tie graduates off the LLM. Cost is the axis this study measures; latency and modality are first-class and default to unconstrained.
- **Tie-to-graduate**: ML graduates once it *ties* the LLM (non-inferiority), not once it beats it.
- **CostToleranceGate**: the non-inferiority gate; ϵ is operator-tunable.
- **The router is permanent**: it holds the rule floor, watches for drift, and re-escalates to the LLM when the data demands — only the LLM *dependency* shrinks.

4.1 Measured latency + cost (the steady-state axes)

Per-call latency (p50): rule 0.003 ms, classical-ML 0.1 ms, MiniLM-embed ML 6-23 ms, LLM 530-570 ms — the rule is ~200,000x faster than the LLM, classical ML ~5,000x. LLM per-call cost \$0.0001-0.0006 (scales with label-prompt length); rule/ML ~\$0. Under a real-time SLO (e.g. 50 ms) the LLM is structurally excluded, so latency *forces* graduation to on-device tiers (latency_profile.json).

4.2 AGA vs the fixed ladder

AGA matches accuracy at far lower labeled-outcome cost:

dataset	oracle	AGA(LODO)	AGA_acc	fixed_acc	AGA_cost	fix_cost
ag_news	ml	model_zeroshot	0.853	0.895	0	5000
atis	ml	ml	0.848	0.848	1000	1000
banking77	ml	model_fewshot	0.807	0.870	40	5000
cifar10	model_zeroshot	ml	0.327	0.327	1000	1000
clinc150	ml	ml	0.838	0.838	5000	5000
codelangs	ml	ml	0.986	0.986	500	500
dbpedia14	ml	ml	0.968	0.968	10000	10000
emotion	ml	model_zeroshot	0.632	0.818	0	5000
esc50	ml	model_fewshot	0.050	0.283	8	100
hwu64	model_fewshot	ml	0.805	0.805	5000	5000

imdb	model_fewshot	model_fewshot	0.958	0.853	40	5000
rotten_tomatoes	model_fewshot	model_zeroshot	0.930	0.723	0	5000
sms_spam	model_fewshot	ml	0.945	0.945	2000	2000
snips	ml	ml	0.983	0.983	500	500
sst2	model_fewshot	model_zeroshot	0.968	0.770	0	5000
trec6	ml	model_zeroshot	0.795	0.855	0	5000
tweet_emotion	model_zeroshot	ml	0.613	0.613	2000	2000
tweet_hate	model_zeroshot	model_fewshot	0.640	0.560	40	250
tweet_offensive	ml	model_fewshot	0.730	0.775	40	2000
twenty_newsgroups	ml	ml	0.660	0.660	10000	10000
yahoo_answers	model_zeroshot	ml	0.670	0.670	10000	10000

```
LODO oracle-match: 7/21 = 0.33
Mean accuracy  AGA 0.762 vs fixed-ladder(ML_PRIMARY) 0.764
Mean labeled outcomes  AGA 2246 vs fixed-ladder 4017
AGA keeps LLM as terminal (skips ML training): 10/21
AGA picks rule/ML over not-better LLM (avoids perpetual cost): 6/21
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4.3 Honest negative: cross-site prediction does not yet generalize

A gradient-boosted predictor of the best tier from early-observable characteristics scores **below the majority baseline** leave-one-dataset-out at 21 datasets, and its apparent skill swings with test-set noise (0.62→0.41 between n=200 and n=400 before significance-aware labeling). **Conclusion: per-site measured routing is what works today; cross-site prediction needs a much larger fleet.** This is a finding, not a failure — it tells practitioners to measure, not guess.

5 Preliminary: image and audio

The gate is modality-agnostic (it consumes (decision, outcome) pairs); only the LLM input path is modality-specific. Rule+ML transfer to CIFAR-10 and ESC-50. The MODEL tier is preliminary: Claude vision on CIFAR images reaches 0.78 zero-shot (but our pixel-logreg ML is a weak baseline), and a spectrogram→vision proxy for ESC-50 **fails** (0.017, below chance) — native-audio models are needed. We report these as preliminary evidence of modality-generality, not as headline results.

6 Limitations

- **Classical-ML baselines are the cheap day-N head** (TF-IDF / pixel / MFCC + logistic regression). **Robustness check (Threat-1 ablation):** a much stronger classical tier — frozen sentence-transformer embeddings (all-MiniLM-L6-v2) + logistic regression — was run on a representative text subset. The "LLM wins on short-text sentiment/moderation" result *survives*: sst2 strong-ML 0.80 vs LLM 0.98; rotten_tomatoes 0.74 vs 0.94; tweet_hate 0.53 vs 0.77 — so it is **not** a weak-baseline artifact. High-cardinality intent still favors classical ML (banking77 0.89, clinc150 0.87, both > LLM). Caveat: frozen embeddings are not uniformly stronger (imdb 0.78 < TF-IDF 0.85 due to MiniLM truncating long reviews); a *fine-tuned* transformer ablation is the remaining future work. (Image ML remains a weak pixel-logreg; a CNN/ViT baseline is needed before drawing image conclusions.)
- Latency/cost are measured on a sample (latency_profile.json); LLM latency is network-bound and will vary.
- **Single LLM (Claude Haiku), single prompt** — accuracies are prompt/model sensitive.
- **Multimodal MODEL tier is a proxy** (spectrogram/frames); native audio/video models are future work. Spectrogram-vision is shown to fail on environmental audio.
- **Cross-site meta-prediction does not generalize** at this fleet size (ğ4.2).

- **n=400 test subsets** $\rightarrow$ accuracy CIs $\approx$ ś0.04; we use significance-aware tie bands accordingly.

7 Related work

FrugalGPT / LLM cascades (cost routing); learning-to-defer (per-example deferral); champion-challenger + shadow deployment (MLOps); McNemar / Dietterich. AGA differs by choosing the *terminal* tier per stream from characteristics and graduating off the LLM under a non-inferiority gate — not per-example routing or a hand-tuned cascade.

8 Reproducibility

One command (`scripts/reproduce_dmlr.sh`) regenerates every number; MODEL stages auto-skip without an API key. See REPRODUCE.md.